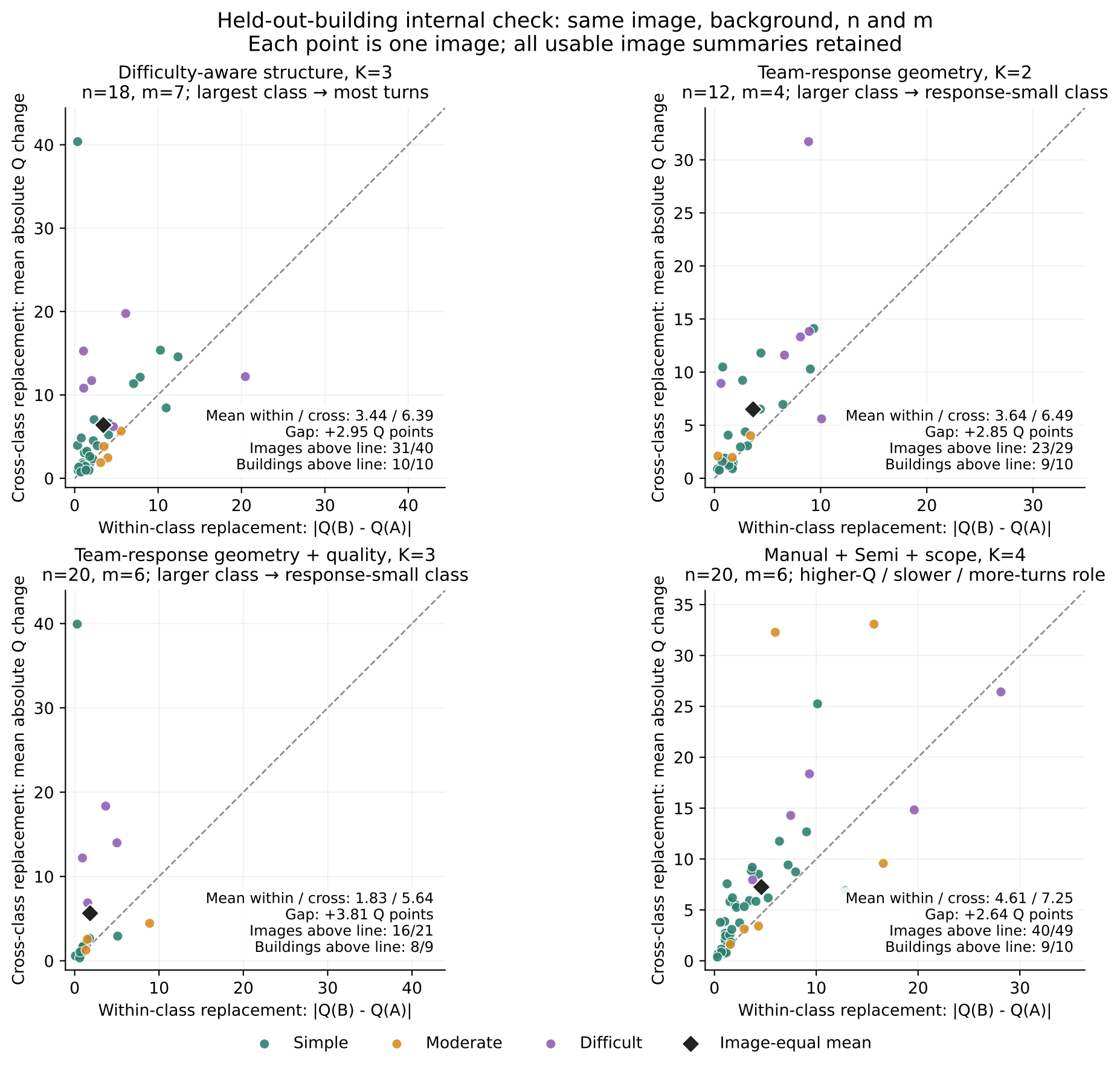
Held-out-building internal check: same image, background, n and m
Each point is one image; all usable image summaries retained
Difficulty-aware structure, K=3
n=18, m=7; largest class → most turns
Mean within / cross: 3.44 / 6.39
Gap: +2.95 Q points
Images above line: 31/40
Buildings above line: 10/10
Team-response geometry, K=2
n=12, m=4; larger class → response-small class
Mean within / cross: 3.64 / 6.49
Gap: +2.85 Q points
Images above line: 23/29
Buildings above line: 9/10
Team-response geometry + quality, K=3
n=20, m=6; larger class → response-small class
Mean within / cross: 1.83 / 5.64
Gap: +3.81 Q points
Images above line: 16/21
Buildings above line: 8/9
Manual + Semi + scope, K=4
n=20, m=6; higher-Q / slower / more-turns role
Mean within / cross: 4.61 / 7.25
Gap: +2.64 Q points
Images above line: 40/49
Buildings above line: 9/10
Cross-class replacement: mean absolute Q change
Within-class replacement: |Q(B) - Q(A)|
Simple
Moderate
Difficult
Image-equal mean